

Rishabh Rudani

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Robotics Engineering Student — Space Robotics — Autonomous Systems — AI Perception — Embedded Robotics

Professional Summary

Robotics Engineering student specializing in **deep learning, semantic segmentation, point cloud data processing, 3D machine vision, aerial drone perception, localization, mapping, robotics simulation, and real-hardware deployment**. Hands-on experience developing ROS 2 autonomy stacks and AI perception models using **PyTorch, TensorFlow, CUDA, CNNs, U-Net, and LSTM networks**. Currently developing **Odysseus Rover**, a space robotics platform with the University of Würzburg in collaboration with **AMADEE-27**, focused on simulation validation, hardware-software co-design, and future real-hardware deployment.

Work & Project Experience

Space Robotics Hardware & Software Developer – Odysseus Rover Dec 2025 – Present
University of Würzburg Collaboration — AMADEE-27 Space Robotics Platform Germany

- Developing a ROS 2-based **planetary rover platform** for autonomous exploration, rock mapping, localization, perception, navigation, and mission execution.
- Building simulation environments to test rover behavior, mapping pipelines, sensor logic, waypoint missions, and autonomous inspection workflows before real-hardware deployment.
- Designing the autonomy architecture across **sensor drivers, filtering, SLAM, localization, Nav2 navigation, visual odometry fallback, rock inspection, and control execution**.
- Integrating **LiDAR, depth camera, IMU, UWB sensors, wheel encoders, point clouds, occupancy grids, and AI-based perception** for robust rover autonomy.
- Supporting hardware-software co-design for UWB anchor deployment, microcontroller-based nodes, battery/solar-powered modules, base-station infrastructure, and embedded system integration.

Robotics Software Engineer Intern Sep 2025 – Apr 2026
Lemvos Robotics GmbH – Autonomous Navigation for Unmanned Surface Vehicle Augsburg, Germany

- Engineered ROS 2 software for an autonomous marine robot covering **localization, mapping, Nav2 navigation, perception, control, mission execution, and runtime debugging**.
- Implemented **GNSS/IMU sensor fusion** with `robot_localization` and developed Nav2-based path-following, command normalization, and hardware-aware control logic.
- Built perception and mapping workflows using computer vision, inverse perspective mapping, bird's-eye-view processing, point-cloud conversion, and local costmap integration.
- Developed Behavior Tree-based mission logic for waypoint execution, hold-position behavior, docking, undocking, safe-stop actions, and operator-facing service interfaces.
- Created validation tools for cross-track error analysis, path-following metrics, telemetry monitoring, runtime diagnostics, and full-stack system testing.

Selected AI & Robotics Projects

U-Net Semantic Segmentation for Water, Ships & Land *PyTorch, CUDA, Deep Learning, Computer Vision*
Built a U-Net-based semantic segmentation model using PyTorch to detect and segment water, ships, and land regions from image data; used CUDA-based GPU acceleration for faster model training and inference.

LSTM Movie Review Sentiment Classification *Python, LSTM, NLP, Deep Learning*
Developed an LSTM-based neural network for movie review sentiment classification using text preprocessing, tokenization, embedding layers, sequence modeling, and recurrent neural network training.

Autonomous Car Simulation *ROS 2 Jazzy, Nav2, LiDAR-SLAM, Gazebo, RViz*
Built a complete autonomous vehicle simulation with LiDAR-based SLAM, local costmap generation, navigation, controller integration, and real-time visualization.

Technical Skills

Robotics & Simulation: ROS 2, Nav2, Gazebo, RViz, URDF, SLAM Toolbox, BehaviorTree.CPP, RoboDK, robotic control, localization, mapping, autonomous navigation, sensor integration

AI, Perception & Machine Vision: Deep learning, PyTorch, TensorFlow, CUDA, CNNs, U-Net, LSTM, NLP, point cloud processing, 3D machine vision, OpenCV, object detection, semantic segmentation, rock mapping, RGB-D perception, aerial drone perception, IPM, BEV

Programming: C++, Python, Java, JavaScript, MATLAB, HTML, CSS

Hardware & Embedded Systems: LiDAR, RGB-D cameras, IMU, GNSS, UWB sensors, wheel encoders, Jetson Orin, CAN/CANopen, actuator control, robotics prototyping, hardware-software integration

DevOps & Tools: Linux, Docker, Git, GitHub, CI/CD pipelines, automated testing, version control, software deployment workflows, Foxglove, Grafana, MATLAB Simulink

Education

B.Sc. Robotics Engineering 2023 – Present
Technical University of Applied Sciences Würzburg-Schweinfurt (THWS) Germany

Relevant coursework: Autonomous Systems, Control Systems, Computer Vision, Machine Learning, Digital Signal Processing, Distributed Systems, Robotics Software Engineering.

Languages

English: Mother tongue / Fluent

German: Intermediate